

HYPONATREMIA

CRITICAL LOW: not stated; LOW <135

SODIUM

RANGE: - 135-145 mEq/L

HYPERNATREMIA

CRITICAL HIGH not stated; >145

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

Sodium mainly affects fluid balance, blood circulation, the brain, nerves, and muscles.

It helps control how much water stays in the bloodstream, supporting blood volume and blood pressure.

It also helps nerve and muscle cells send electrical signals.

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Na loss: vomiting, NG suction, sweating

- Diuretics / renal loss; low intake or NPO

- CHF or hypotonic IV fluid -> dilution

- SIADH -> excess ADH -> water retention

Signs & Symptoms when it's LOW? Critically LOW?

- Confusion, lethargy, poor concentration

- Seizures or stupor = severe neurologic danger

- Weakness; decreased deep tendon reflexes (DTR); shallow breathing can be late

- Abdominal cramps/low BS; low BP and urine output

How do we fix it?

- Trend Na; neuro/VS/RR/I&O; seizure precautions

- SIADH/dilution: fluid restrict +/- diuretic as ordered

- Hypertonic IV as ordered; watch fluid overload

- Treat cause; moderate salty foods only if appropriate

What causes it to INCREASE?

- Water loss: fever, sweating, vomiting, diarrhea

- Diabetes insipidus -> free-water loss

- Excess Na: oral/IV, tube feed, hypertonic fluid

- Corticosteroids / aldosterone excess

Signs & Symptoms when it's HIGH? Critically HIGH?

- Thirst; dry mouth/skin; fever or flushed skin

- Agitation or confusion

- Edema / fluid retention when Na gain is the cause

- Low urine output; dehydration findings with water loss

How do we fix it?

- Trend Na; neuro/VS/I&O/daily wt; fall precautions

- Ordered isotonic or hypotonic IV fluid

- Correct slowly -> rapid drop risks cerebral edema

- Restrict Na; treat cause (example: DI)

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Soy sauce, canned soups, deli cold cuts, frozen "tv" dinners, fried foods, pickles, bacon,

LOW: ENCOURAGE sources above only if appropriate

HIGH: LIMIT / AVOID sources above

(arranged from higher content to lower but all rich in sodium.)

HYPOKALEMIA

CRITICAL LOW: not stated; LOW <3.5; <2.5 urgent

POTASSIUM

RANGE: - 3.5-5.0 mmol/L

HYPERKALEMIA

CRITICAL HIGH not stated; >5.0

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

Potassium mainly affects the heart, nerves, muscles, and digestive tract.

It helps nerves send signals, muscles contract, and the heart beat steadily.

The kidneys keep potassium balanced by removing extra potassium in the urine.

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- GI loss: vomiting, diarrhea, NG drainage
- Loop/thiazide diuretics; poor intake/dehydration
- Insulin or albuterol -> K moves into cells
- Hyperaldosteronism / corticosteroids

Signs & Symptoms when it's LOW? Critically LOW?

- Weak/irregular pulse; low BP or orthostasis
- Weakness/cramps; decreased deep tendon reflexes (DTR); paralysis if severe
- Shallow RR; diminished breath sounds; low bowel sounds
- ECG: ST depression, inverted T, prominent U wave

How do we fix it?

- Telemetry; monitor VS/RR/GI/DTR; check K + Mg
- Check renal function + urine output before replacement
- Oral K with food when possible
- IV K: dilute/pump/slow; monitor site - NEVER IV push

What causes it to INCREASE?

- Renal impairment -> decreased K excretion
- ACE inhibitor, spironolactone, or NSAID
- Addison disease -> too little aldosterone
- Burns/trauma (cell lysis) or excess IV K

Signs & Symptoms when it's HIGH? Critically HIGH?

- Irregular/brady pulse; low BP; VF if severe
- ECG: peaked T waves and ST elevation
- Weakness/cramps; tingling; decreased deep tendon reflexes (DTR)
- Respiratory failure; hyperactive BS/diarrhea; confusion

How do we fix it?

- Continuous ECG; stop K; question K-retaining meds
- IV calcium gluconate -> protects the heart
- Insulin+dextrose / albuterol / bicarb -> shift K in
- Diuretic, K binder, or dialysis -> removes K

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Baked potato, spinach, avocado, white beans, sweet potato, tomato sauce, banana, orange juice, lentils

LOW: ENCOURAGE potassium foods above

HIGH: LIMIT foods; NO salt substitutes

(arranged from higher content to lower but all rich in potassium.)

HYPOCALCEMIA

CRITICAL LOW: not stated; LOW <9.0

CALCIUM

RANGE: - 9-11 mg/dL*

HYPERCALCEMIA

CRITICAL HIGH not stated; >11

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

Calcium mainly affects bones, muscles, nerves, the heart, and blood clotting.

It strengthens bones and teeth and helps muscles contract, nerves signal, the heart beat, and blood clot.

Parathyroid hormone (PTH), vitamin D, and the kidneys work together to keep calcium balanced.

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Hypoparathyroidism or thyroidectomy
- Pancreatitis; low vitamin D or magnesium
- High phosphate (including phosphate enemas)
- Loop diuretics/laxatives or chronic GI/renal disease

Signs & Symptoms when it's LOW? Critically LOW?

- Trousseau: hand/arm spasm when BP cuff is inflated
- Chvostek: facial twitch when cheek/temple is tapped
- Cramps/tetany; laryngospasm/dyspnea = airway danger
- Confusion/seizures; ECG: prolonged QT/ST; slow clotting/bleeding

How do we fix it?

- ECG + airway/neuro; seizure/fall precautions
- IV calcium for severe symptoms as ordered
- Oral calcium + vitamin D when stable
- Monitor Ca/Mg/phosphate; encourage calcium foods

What causes it to INCREASE?

- Hyperparathyroidism
- Malignancy -> calcium released from bone
- Excess calcium antacids / supplements
- Low phosphate

Signs & Symptoms when it's HIGH? Critically HIGH?

- Low HR/RR/BP; weak respirations
- ECG: short QT; widened T wave
- Constipation, N&V; renal stones; bone pain
- Decreased deep tendon reflexes (DTR); weakness; confusion / altered alertness

How do we fix it?

- IV normal saline; monitor urine output + renal function
- Loop diuretic after hydration as ordered
- Monitor ECG/neuro; fall precautions
- Calcitonin as ordered; dialysis if severe

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Yogurt, milk, sardines w/bones, cheese, cottage cheese, cooked kale, almonds, broccoli, white beans, ice cream.

LOW: ENCOURAGE calcium foods above

HIGH: LIMIT sources above

(arranged from higher content to lower but all rich in calcium.)

HYPOMAGNESEMIA

CRITICAL LOW: not stated; LOW <1.5

MAGNESIUM

RANGE: - 1.5-2.5 mEq/L

HYPERMAGNESEMIA

CRITICAL HIGH not stated; >2.5

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

Magnesium mainly affects the muscles, nerves, heart, and bones.

It helps muscles relax, supports normal nerve function, and keeps the heartbeat steady.

It works closely with calcium and potassium, so these levels are often checked together.

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Alcohol misuse -> poor GI absorption

- NG suction / N&V / diarrhea; diuretics

- Aminoglycoside antibiotics

- Pregnancy-related malnutrition

Signs & Symptoms when it's LOW? Critically LOW?

- Tachycardia/tachypnea; dyspnea

- ECG: long QT, depressed ST, inverted T

- Increased deep tendon reflexes (DTR/hyperreflexia); clonus; tingling

- Confusion/seizures; torsades or VF = danger

How do we fix it?

- ECG; seizure precautions; assess RR + swallowing

- Check Mg/Ca/K and DTR/clonus

- IV magnesium sulfate slowly; recheck labs

- Treat cause; encourage Mg-rich foods

What causes it to INCREASE?

- Renal failure -> decreased Mg excretion

- Excess Mg antacids or laxatives

- Diabetic ketoacidosis

- Addison disease / hyperkalemia

Signs & Symptoms when it's HIGH? Critically HIGH?

- Low BP, RR, and HR

- ECG: widened QRS and prolonged PR

- Decreased or absent deep tendon reflexes (DTR); weakness; low bowel sounds

- Drowsy/lethargic; coma if severe

How do we fix it?

- Stop Mg products; monitor VS/ECG/RR/LOC/DTR

- IV calcium gluconate as ordered

- Fluids + loop diuretic if renal function allows

- Hemodialysis if severe or renal failure

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Pumpkin seeds, almonds cooked spinach, black beans, peanut butter, whole-grain cereal, brown rice, oatmeal, avocado

LOW: ENCOURAGE magnesium foods above

HIGH: LIMIT foods + Mg antacids/laxatives

(arranged from higher content to lower but all rich in magnesium.)

HYPOPHOSPHATEMIA

CRITICAL LOW: not stated; LOW <2.5

PHOSPHORUS

RANGE: - 2.5-4.5 mg/dL

HYPERPHOSPHATEMIA

CRITICAL HIGH not stated; >4.5

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?
* Note: Understanding function helps understand the problems when it is low/high *

Phosphorus mainly affects bones, muscles, and cells that need energy.

It works with calcium to build strong bones and teeth, and cells use it to make energy.

The kidneys and parathyroid glands regulate phosphorus; it usually moves opposite calcium.

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Antacids/malabsorption or low vitamin D
- Hyperparathyroidism / hypercalcemia
- Severe malnutrition or excess alcohol
- Burns, hyperglycemia, or malignancy-related renal loss

Signs & Symptoms when it's LOW? Critically LOW?

- Muscle/respiratory weakness; decreased deep tendon reflexes (DTR)
- Decreased cardiac output
- Osteomalacia -> fracture/fall risk
- Confusion/seizures; low platelets -> bleeding

How do we fix it?

- Check phosphate/Ca, renal function, ECG + respirations
- Oral phosphate + vitamin D as ordered
- IV phosphate if severe; confirm kidney function
- Monitor calcium; fall/seizure precautions; diet

What causes it to INCREASE?

- Renal insufficiency/failure -> poor excretion
- Fleet / sodium-phosphate enema
- Excess vitamin D or hypoparathyroidism
- Chemotherapy / cell lysis

Signs & Symptoms when it's HIGH? Critically HIGH?

- Trousseau: hand/arm spasm when BP cuff is inflated
- Chvostek: facial twitch when cheek/temple is tapped
- Increased deep tendon reflexes (DTR); tetany/spasms; bone pain
- Laryngospasm = airway danger; confusion

How do we fix it?

- Trend phosphate/Ca; ECG/neuro/airway monitoring
- Calcium acetate (PhosLo) WITH FOOD
- Avoid phosphate enemas; limit phosphate foods
- Dialysis if severe or renal failure

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Pumpkin seeds, salmon, chicken breast, pork chops, Greek yogurt, lentils, milk, peanut butter, black beans, cheese, eggs

LOW: ENCOURAGE phosphate foods above

HIGH: LIMIT sources above

HYPOCHLOREMIA

CRITICAL LOW: not stated; LOW <98

CHLORIDE (Cl-)

HYPERCHLOREMIA

CRITICAL HIGH not stated; >107

RANGE: 98-107 mEq/L*

What does it do in the body? What body system/organ(s) favors it? What is the expected function for that body system(s)?

* Note: Understanding function helps understand the problems when it is low/high *

Chloride mainly affects body-fluid balance, the stomach, and acid-base balance.

It follows sodium to balance body fluids and forms stomach acid for digestion.

The kidneys regulate chloride, which works with bicarbonate to keep blood pH normal.

IMPLICATIONS & NURSING MANAGEMENT

What causes it to DECREASE?

- Vomiting/NG suction -> loss of gastric HCl
- Loop/thiazide diuretics -> renal chloride loss
- Sweating/burns or excess water/HF/SIADH
- Metabolic alkalosis (bicarbonate rises)

Signs & Symptoms when it's LOW? Critically LOW?

- Usually reflects alkalosis or fluid loss
- Weakness/fatigue; cramps/twitching; tingling
- Confusion/irritability; slow or shallow respirations
- Dysrhythmia/tetany/seizure if K or Ca also low

How do we fix it?

- Assess cause + fluid status; VS/neuro/RR/I&O
- Trend Cl/Na/K/HCO₃; ABG if symptomatic
- Volume loss: NS; dilution: fluid restrict as ordered
- Low K: KCl as ordered after renal/urine check

What causes it to INCREASE?

- Large 0.9% saline / excess chloride load
- Severe diarrhea -> bicarbonate loss
- Dehydration/free-water loss: fever, sweating, DI
- Renal dysfunction/RTA or metabolic acidosis

Signs & Symptoms when it's HIGH? Critically HIGH?

- Usually reflects dehydration or metabolic acidosis
- Thirst/dry mucosa; weakness/fatigue; confusion
- Deep, rapid respirations with acidosis
- Low BP/tachycardia/oliguria; edema if fluid overloaded

How do we fix it?

- Trend Cl/Na/K/HCO₃/renal labs; assess lungs/I&O
- Stop/reassess chloride-rich IV; use ordered balanced fluid
- Correct water/volume deficit with ordered fluids
- Treat diarrhea, renal cause, or acidosis

ENCOURAGE

DIETARY SOURCES (HIGHEST CONTENT PER SERVING)

AVOID

Soy sauce, canned soups, deli cold cuts, frozen "tv" dinners, fried foods, pickles, bacon,

LOW: ENCOURAGE salty foods only if prescribed

HIGH: LIMIT table salt + processed foods

(chloride is usually consumed with sodium as table salt.)